Fixed-margin prediction of RNA-protein association across cohorts

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Abstract

Background: Linked single-cell assays resolve cross-modal dependence because they measure molecular states in the same cell. Recipient cohorts may provide the marginal state frequencies of both assays without preserving cell-level pairing. Copying a source joint distribution then confounds association with cohort-specific abundance.

Results: We predict the missing recipient joint table by estimating its centered log-linear interaction at fixed margins. Penalized exact conditional likelihood estimates the population interaction and shrinks donor deviations; the estimate is then combined with each recipient's observed margins. Fitted on 36 Cambridge donors, the method achieved mean deviance per cell 0.01220 across 56 Newcastle donors, compared with 0.01478 for signed-root deviance-statistic transfer. The reduction was 17.46% (paired-donor 95% confidence interval for the loss difference, -0.00413 to -0.00080), with lower loss in 50/56 donors. Mean deviance for the destroyed-link control was 0.02274, and the intact-link method had lower loss in 51/56 donors. In fitted-baseline analyses, the estimator reduced deviance by 6.18% relative to the common-effect conditional estimate. Across 128 ground-truth simulations, fixed-margin reconstruction absorbed recipient abundance shift, and donor shrinkage improved prediction when source interactions were heterogeneous.

Conclusions: Fixed-margin prediction separates cross-modal association from cohort-specific abundance. Across held donors and ground-truth simulations, recipient margins corrected abundance shift, whereas donor shrinkage helped when source interactions were heterogeneous.

Keywords: single-cell multi-omics; conditional likelihood; log-linear association; transfer learning; CITE-seq; reproducibility

1 Background

Linked single-cell assays measure RNA, chromatin, proteins, molecular age, and lineage in the same cell or experimentally related unit [1–6]. Joint latent-variable models integrate paired measurements or predict a missing modality. We ask whether cross-modal association learned from linked source donors can predict a recipient joint distribution from its two observed margins.

Marginal frequencies change with cell composition, activation, sequencing depth, and assay scale. Copying a source table therefore transfers abundance together with dependence. Signed Pearson and likelihood-ratio statistics summarize departure from independence, but their scale and attainable range depend on the margins. A statistic pooled across donors can consequently change meaning at recipient margins.

Log-linear interactions isolate association in contingency tables [7–10]. Conditional likelihood removes row and column nuisance parameters, and common-effect or mixed-effects models share odds-ratio information across observed strata [11–13]. Copula and differential-association methods also estimate dependence from observed single-cell or multi-omic pairings [14–17]. These methods estimate dependence where pairings are observed; our target is the unobserved joint table at new margins.

Centered log-linear interactions and conditional odds-ratio estimation are established tools for observed contingency tables. We use them to predict an unobserved recipient table from linked source donors and the recipient margins. The method pools heterogeneous donor effects, then reconstructs each table at the abundance observed in its recipient. We prove that the penalized fit is unique and bound recipient risk by source estimation error and recipient drift. A prespecified CITE-seq held-site experiment and ground-truth simulations compare the method with signed-statistic transfer, fitted interaction models, and margin-preserving link destruction.

2 Methods

2.1 Association coordinates

For ordered marker pair e, let P_e be a positive joint-probability table for linked states U and V. Orthonormal contrast matrices B_U and B_V remove the row and column components of its log table. We call the resulting centered log-linear interaction the *coupling field*:

$$\Theta(P_e) = B_U^{\mathsf{T}} \log(P_e) B_V \tag{1}$$

where the logarithm is elementwise. This field contains all log-linear interaction degrees of freedom, is invariant to positive row or column tilts, and vanishes exactly at independence.

For a binary table, Eq. (1) is one-half of the log odds ratio,

$$\Theta(P_e) = \frac{1}{2} \log \frac{P_{e,00} P_{e,11}}{P_{e,01} P_{e,10}} = \frac{\theta_e}{2}. \tag{2}$$

Proposition S1 in Additional file 1 proves these properties.

2.2 Hierarchical conditional estimator

Let A_{de} be the upper-left count in the binary table for donor d and marker pair e, and let m_{de} denote its margins. Conditional on these margins, the table has one free count. Its exact distribution is

$$p(a \mid m_{de}, \theta) = \frac{w_{de}(a) \exp(\theta a)}{\sum_{b \in \mathcal{A}_{de}} w_{de}(b) \exp(\theta b)}, \tag{3}$$

where $w_{de}(a)$ is the central-hypergeometric combinatorial weight. The distribution depends on the log odds θ but not on row or column effects.

We write the donor-specific log odds as a population effect μ_e plus a donor deviation δ_{de}. The estimator minimizes

$$\begin{aligned} \mathcal{L}(\mu, \delta) = &-\frac{1}{D} \sum_{d,e} \log p(A_{de} \mid m_{de}, \mu_e + \delta_{de}) \\ &+ \frac{\gamma}{2} \sum_{d,e} \delta_{de}^2 + \frac{\rho}{2} \|\mu\|_2^2. \end{aligned} \tag{4}$$

The first term is the donor-normalized negative exact conditional log likelihood summed over marker pairs. The panel objective is composite because marker pairs from one donor reuse cells. The parameter γ shrinks donor deviations toward the population interaction, and ρ penalizes the population interaction. Penalty scaling, optimization, and convergence criteria are given in Additional file 1.

Conditioning a Poisson log-linear model on both margins removes every row and column parameter and yields Eq. (3). Positive donor and population penalties give Eq. (4) a unique finite minimizer; Proposition S2 in Additional file 1 gives the derivation and proof.

For recipient q, the prediction is the fixed-margin expected table at transported population log odds,

$$\widehat{N}_{qe} = \mathbb{E}_{\alpha \widehat{\mu}_e}[N_{qe} \mid m_{qe}], \tag{5}$$

where the development design selects α from a prespecified study grid. The binary cohort studies used $\{0.5, 0.75, 1, 1.25\}$, and the longitudinal calibration also included zero. This table preserves the recipient margins without a pseudocount or post-fit projection.

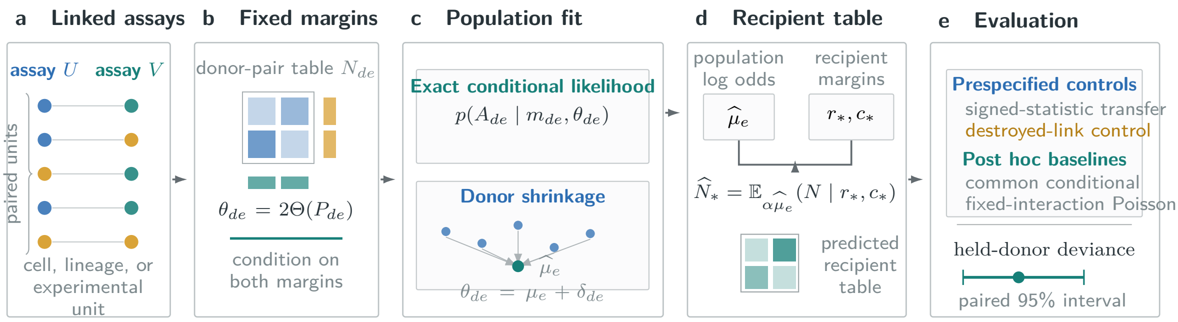

Fig. 1 Conditional transport of pairwise association. Linked measurements define donor-level joint tables. Conditioning on both margins isolates log-linear association, and donor shrinkage estimates the population component shared across donors. Combining this estimate with recipient margins reconstructs a feasible recipient table. Held comparisons use signed-statistic transfer and margin-preserving link destruction. Common-effect conditional and fixed-interaction Poisson baselines were evaluated post hoc.

Proposition S3 bounds expected excess conditional log loss by fixed-margin Fisher information times the sum of squared source transport error and recipient drift variance. Source data can reduce the first term, whereas recipient drift determines the residual floor. Additional file 1 gives the proof and a finite-support upper bound. We assessed held reconstructed tables separately by multinomial deviance.

2.3 Classical comparisons and controls

The matched statistic comparator transfers either the signed root Pearson statistic or the signed root likelihood-ratio deviance from the row-plus-column independence model. We normalized statistics by table size, pooled them across source donors, rescaled them for each recipient, and inverted them at its margins. Each study fixed its pooling rule in advance; development data selected the statistic, any evaluated null centering, and the transport multiplier. Both methods received the same recipient margins without recipient pairings.

Two fitted interaction baselines compared alternative pooling rules. The common-effect conditional maximum-likelihood estimate fitted one unpenalized log odds per marker pair across donor strata. The pooled-table baseline used the sample log odds ratio of the donor-pooled table. In the post hoc comparison, both coefficients were reconstructed by fixed-margin conditional expectation with a development-selected transport multiplier. We call the second method pooled-table log odds with conditional reconstruction.

A separate classical baseline treated the pooled-table coefficient as a fixed interaction in a saturated Poisson log-linear model. After development-only selection of the transport multiplier, we refitted recipient row and column parameters at that interaction. The resulting continuous table has the recipient margins and transported odds ratio. This construction differs from the conditional noncentral-hypergeometric expectation used above.

The destroyed-link control permuted complete protein profiles within each donor before joint tables were formed. This preserved RNA and protein margins and dependence within the protein panel but removed cell-level RNA-protein linkage. We refitted the selected conditional estimator to the permuted tables. Fixed-margin independence provided a further control.

For observed table T, positive prediction $\widehat{T}$, and table total n, the loss is multinomial deviance per cell,

$$D(T, \widehat{T}) = \frac{2}{n} \sum_{i,j:T_{ij}>0} T_{ij} \log \frac{T_{ij}}{\widehat{T}_{ij}}. \qquad (6)$$

Entity losses were averaged within each evaluation unit. Confirmatory means weighted physical donors equally. The 95% intervals used 20,000 paired bootstrap draws at the donor or prespecified batch-block level, conditional on the source fit, selected configuration, and deterministic cell sample. Development intervals resampled the displayed samples or batches.

2.4 Ground-truth simulation

We simulated 128 replicates from the exact fixed-margin model with known population log odds, 12 source donors, 12 recipient donors, 16 marker pairs, and 128 observations per table. Source donor effects were either homogeneous or normally heterogeneous with standard deviation 1.2. Recipient margins either followed the source distribution or underwent a strong shift. Hyperparameters and transport multipliers were fixed across all four scenarios. We compared the hierarchical estimator with the common-effect conditional estimate and raw and null-centered signed-root deviance-statistic transfer. Paired bootstrap intervals resampled simulation replicates in 20,000 draws.

2.5 Prespecified held evaluation

The Stephenson held-site confirmation followed an executable public plan released before the linked evaluation outcomes were accessed. It fixed eligibility, donor allocation, cell selection, state definitions, marker panels, estimator grids, comparators, loss, uncertainty, and success criteria. After the pilot passed, all development donors supplied the final fit. Predictions from the recipient margins and their checksums were published before the recipient RNA-protein pairings were accessed.

Confirmation required at least 5% lower mean deviance than both the selected statistic-transfer and destroyed-link controls, a paired-bootstrap upper endpoint below zero, and a one-sided sign-test $P \leq 0.025$. No configuration was revised after release of the plan.

2.6 Datasets and state construction

Stephenson et al. profiled PBMC RNA and surface proteins with a common TotalSeq-C panel across Cambridge and Newcastle [18]. The split used 12 Cambridge calibration donors, 24 Cambridge pilot donors, and 56 Newcastle donors for the held-site test. Nine cognate RNA and protein markers defined 81 ordered pairs.

Each donor contributed 512 cells selected by a deterministic rule that did not access RNA or ADT counts. RNA state recorded raw detection. Deterministic within-donor protein ranks assigned 256 low and 256 high cells, fixing every protein margin before evaluation. Additional file 1 reports the secondary public-study sequence and its evidence roles.

2.7 Software and reproducibility

The tagged public release contains the estimator and cohort reducers, the analysis plans and held predictions, the derived cohort results, the programs for the ground-truth simulation and risk stress test, and verification tests. OpenAI Codex assisted with code development, literature retrieval, manuscript editing, and LaTeX preparation.

3 Results

3.1 Association transferred across recruitment sites

Across 56 Newcastle donors, the hierarchical conditional estimator reached mean deviance 0.01220, compared with 0.01478 for signed-root deviance-statistic transfer. The corresponding 17.46% reduction had a paired-donor loss-difference interval of -0.00413 to -0.00080. The conditional estimate was better in 50/56 donors ($P = 5.09\times10^{-10}$, one-sided sign test). All 36 Cambridge development donors supplied the final source fit after pilot selection.

The estimator also reduced deviance by 46.36% relative to the destroyed-link fit of 0.02274 (paired-donor interval -0.01323 to -0.00749; 51/56 donors favorable). In post hoc fitted-baseline analyses, it improved on the common-effect conditional estimate by 6.18% (relative-reduction interval 3.84% to 8.47%) and on pooled-table log odds with conditional reconstruction by 3.39% (1.42% to 5.37%).

3.2 Conditioning and donor shrinkage separated distinct errors

The factorial simulation separated donor heterogeneity from margin shift (Fig. 2). With heterogeneous donor effects and source-like margins, the hierarchical estimator reached deviance 0.01493, compared with 0.01521 for the common-effect conditional estimate. The reduction was 1.85% (95% interval 1.52% to 2.18%). After the recipient margins shifted, conditional transport reduced deviance by 65.49% (63.02% to 67.72%) relative to raw signed-root deviance-statistic transfer.

Table 1 Held-site performance against matched controls and fitted interaction baselines.

Comparator	Analysis role	Comparator loss	Reduction	Difference 95% CI	Favorable donors
Signed-root deviance transfer	prespecified	0.01478	17.46%	-0.00413 to -0.00080	50/56
Destroyed source linkage	prespecified	0.02274	46.36%	-0.01323 to -0.00749	51/56
Common-effect conditional estimate	fitted baseline	0.01300	6.18%	-0.00121 to -0.00046	46/56
Pooled-table log odds	fitted baseline	0.01263	3.39%	-0.00075 to -0.00017	38/56

The hierarchical conditional loss was 0.01220. Differences are hierarchical minus comparator deviance per cell. Intervals use 20,000 paired donor-bootstrap draws. Fitted-baseline comparisons were specified after the held result and are descriptive.

Under homogeneous donor effects and source-like margins, common-effect conditional deviance was 0.008660, compared with 0.008752 for the hierarchical estimator, a shrinkage cost of 1.06%. After margin shift, the two conditional estimators remained close (0.008640 and 0.008674), whereas statistic-transfer deviance rose to 0.02943. Fixed-margin reconstruction therefore absorbed the marginal shift in this model; donor shrinkage contributed when source associations varied across donors.

The finite-support calculations satisfied both risk bounds throughout the prespecified stress grid. Under unbiased transport, risk increased monotonically with recipient drift in every margin family (Additional file 1).

4 Discussion

Recipient margins and source heterogeneity govern different parts of the prediction. Fixed-margin reconstruction absorbed abundance shifts in simulation and transferred RNA-protein association from Cambridge to Newcastle without copying source composition. Donor shrinkage improved on common-effect conditional estimation at the held site and under simulated heterogeneity. Its advantage disappeared when simulated source effects were homogeneous, locating the gain in donor pooling rather than in the interaction coordinate alone.

Classical stratified odds-ratio models estimate association among observed tables. Our target is an unobserved recipient table constrained by its observed margins. Signed-statistic transfer was the prespecified comparator because it uses exactly the available information: linked source tables and unpaired recipient margins. Models such as totalVI and MultiVI solve missing-modality problems from paired recipient observations or cell-level covariates and cannot be fitted under this recipient-unpaired protocol without changing the task. The common-effect conditional estimate provides the closest fitted classical comparison under the same information boundary; hierarchical pooling reduced its held-site deviance by 6.18%.

5 Limitations

The held confirmation is an observational RNA-protein study and does not establish perturbation causality or a molecular mechanism. It uses binary states, 512 cells per donor, and nine RNA and nine protein markers. The 81 pairwise tables within a donor reuse cells and marker vectors. The estimand is conditional dependence, not zero-shot protein abundance. Broader panels, technologies, tissues, state resolutions, and marker sets require independent replication.

The held-site selection chose zero graph penalty, so the empirical claim concerns hierarchical conditional transfer without graph or hypergraph regularization. The fitted-interaction comparisons were specified after the held result and are descriptive. A Stephenson sensitivity retained 29 of 32 repeated destroyed-link fits because three failed the numerical criterion.

The Stephenson sign test is study specific and does not correct familywise error across the preceding public-study search. Public plans prevented outcome-guided reruns within each study, but the sequence selected datasets that passed file, feature, support, and development gates. Additional file

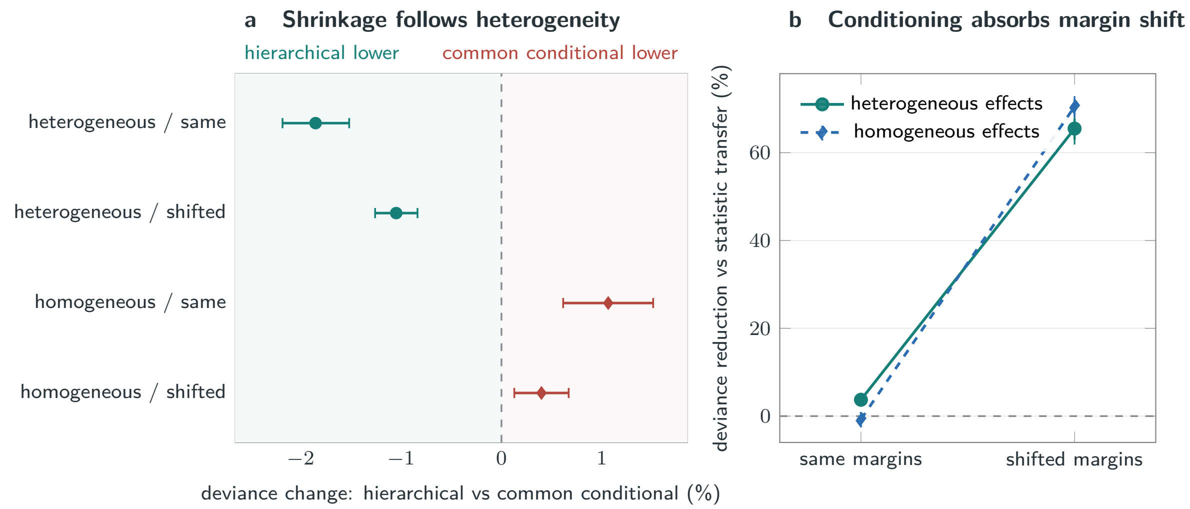

Fig. 2 **Ground-truth estimator behavior.** **a**, relative change in recipient deviance for the hierarchical estimator against the common-effect conditional estimate. Negative values favor hierarchical shrinkage. **b**, reduction in deviance relative to raw signed-root deviance-statistic transfer before and after recipient-margin shift. Bars are 95% paired simulation-replicate bootstrap intervals across 128 replicates.

1 reports the full sequence, including unsuccessful transfers and a post-access corrected cohort analysis.

The exploratory multistate extension depends on discretization, a pseudocount, and finite permutation centering. Spatial applications require linked observations because unpaired regional summaries do not supply the joint tables required by the estimand.

6 Conclusions

Fixed-margin interaction prediction reconstructs cross-modal association at recipient-specific abundance. It outperformed prespecified signed-statistic and destroyed-link controls in a held-site experiment. Recipient margins address abundance shift; source heterogeneity determines the value of donor shrinkage.

Abbreviations

ADT: antibody-derived tag; PBMC: peripheral blood mononuclear cell; RNA: ribonucleic acid.

Declarations

Ethics approval and consent to participate

This study used only public, de-identified data and involved no new participants, specimens, interventions, or identifiable records; the original studies report their ethics approvals and consent procedures.

Consent for publication

Not applicable.

Availability of data and materials

All source datasets are public through the accessions listed in Additional file 1 and retain their original licenses. Analysis plans, MIT-licensed software, cohort and simulation results, study records, and verification scripts are available in the immutable v2.0.3 release at https://github.com/sushaan-k/coupling-fields-benchmark/releases/tag/coupling-fields-v2.0.3-public-benchmark. Raw matrices are not redistributed. The platform-independent software requires Python 3.9 or newer and the version-pinned NumPy and SciPy dependencies in the release; there are no additional restrictions on use.

Competing interests

Competing-interest information is withheld in the blinded manuscript.

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Authors' contributions

Author names and contribution statements are withheld in the blinded manuscript.

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Additional file

Additional file 1 (PDF). *Mathematical results and public benchmark details*. Proofs, algorithms, dataset provenance, study status, comparator definitions, sensitivity analyses, and reproducibility information.

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